

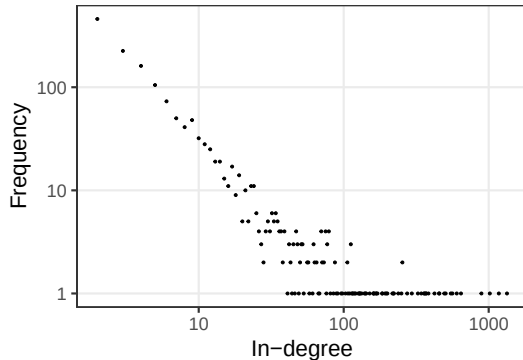
# Evidencing preferential attachment in dependency network evolution

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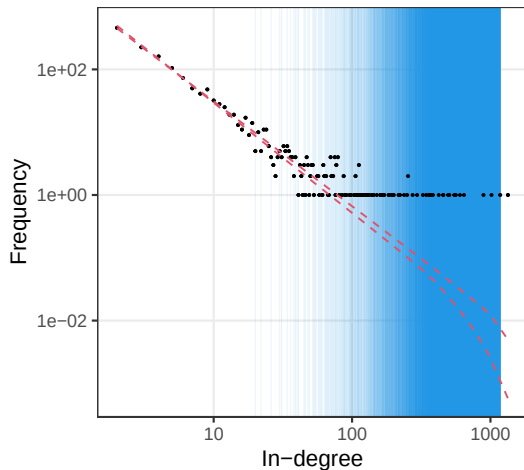
# Introduction

- ▶ Interest in degree distribution
- ▶ Power law  $\rightarrow$  Scale-freeness
- ▶  $\rightarrow$  Preferential attachment
  1. New node joins & brings new edges
  2. Node  $i$  selected with weight  $g(d_i)$
  3. Barabási and Albert (1999):  
 $g(d_i) = d_i$



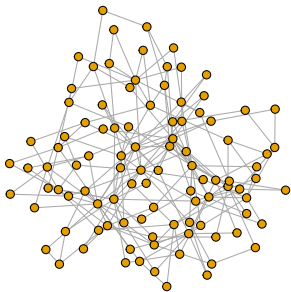
## A few issues

1. General case  $\nrightarrow$  power law
    - ▶  $g(d_i) = d_i^\alpha$  (Krapivsky and Redner 2001)
  2. Other models  $\rightarrow$  scale-freeness
    - ▶ Generalised random graphs (Hofstad 2016)
  3. Debate on ubiquity
    - ▶ Broido and Clauset (2019): rare in real networks
    - ▶ Voitalov et al. (2019): rebuttal
    - ▶ Lee, Eastoe, and Farrell (2024): partial scale-freeness
- ▶ Main cause: only snapshots available

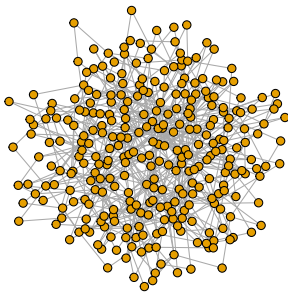


# Take a step back

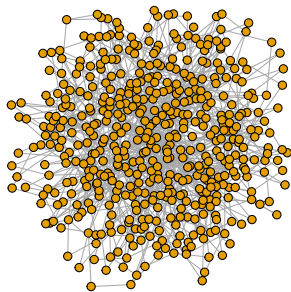
$t-1$



$t$



$t+1$



# Goals

**If** we have the evolution data:

1. Can we evidence the preferential attachment?
2. How do we model the data in a probabilistic / statistical way?
3. What is the precise form of the weight function?
  - ▶  $g(d_i) = d_i^\alpha$  (including  $\alpha = 1$ )?
  - ▶  $g(d_i) =$  a more general form?

## Daily increments of R packages on CRAN

| ##    |  | from           | to      | add   |
|-------|--|----------------|---------|-------|
| ## 1  |  | glmmsr         | methods | FALSE |
| ## 2  |  | GpGp           | FNN     | TRUE  |
| ## 3  |  | pmxTools       | stats   | FALSE |
| ## 4  |  | ceterisParibus | knitr   | TRUE  |
| ## 5  |  | pmxTools       | xpose   | FALSE |
| ## 6  |  | mRchmadness    | shiny   | FALSE |
| ## 7  |  | glmmsr         | lme4    | FALSE |
| ## 8  |  | mRchmadness    | rvest   | FALSE |
| ## 9  |  | xpose          | dplyr   | FALSE |
| ## 10 |  | xpose          | utils   | FALSE |

⋮

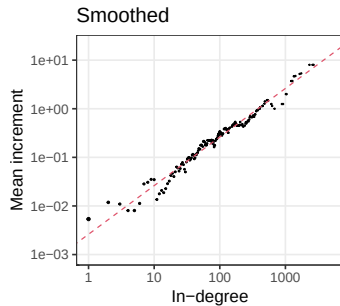
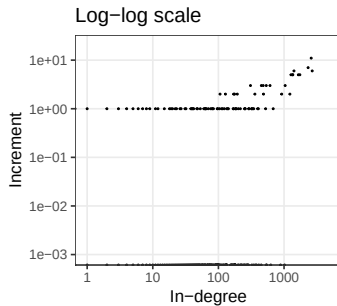
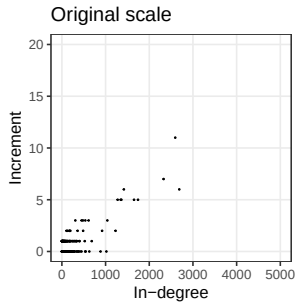
- ▶ Change in Imports for one day
- ▶ Obtained by  
`crandep::get_dep_all_packages()`
- ▶ ... which uses  
`tool::CRAN_package_db()`

## Aggregating

| ##    | previous date | indegree | increment | count |
|-------|---------------|----------|-----------|-------|
| ## 1  | 2019-01-29    | 0        | 0         | 9315  |
| ## 2  | 2019-01-29    | 0        | 1         | 1     |
| ## 3  | 2019-01-29    | 1        | 0         | 1306  |
| ## 4  | 2019-01-29    | 2        | 0         | 459   |
| ## 5  | 2019-01-29    | 3        | 0         | 226   |
| ## 6  | 2019-01-29    | 4        | 0         | 161   |
| ## 7  | 2019-01-29    | 5        | 0         | 103   |
| ## 8  | 2019-01-29    | 6        | 0         | 75    |
| ## 9  | 2019-01-29    | 7        | 0         | 49    |
| ## 10 | 2019-01-29    | 8        | 0         | 40    |

⋮

# Scatter plot



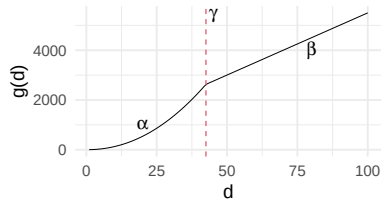


## Regress increment on in-degree

- ▶ The **statistical** model based on preferential attachment
- ▶  $Y_i$ : number of new edges / increments for node  $i$
- ▶  $\{Y_i\} \sim$  Multinomial with prob.  $\left\{ \frac{d_i^\alpha}{\sum_j d_j^\alpha} \right\}$ , 1 new edge
- ▶  $\{Y_i\} | M \sim$  Multinomial with prob.  $\left\{ \frac{d_i^\alpha}{\sum_j d_j^\alpha} \right\}$ ,  $M \sim \text{Poisson}(\mu)$  new edges
  - ▶  $\Rightarrow \{Y_i\} \sim$  Independent Poissons with mean  $\left\{ \frac{\mu d_i^\alpha}{\sum_j d_j^\alpha} \right\}$
  - ▶  $\log E(Y_i) = \alpha \log d_i + \log \mu - \underbrace{\log \left( \sum_j d_j^\alpha \right)}_{\text{constant}}$

# General weight function

- ▶  $g(d) = d^\alpha$ 
  - ▶ Inflexible tail behaviour of degree distribution
  - ▶ Krapivsky and Redner (2001), Rudas, Tóth, and Valkó (2007)
- ▶ Real data displays subtle tail behaviour (Lee, Eastoe, and Farrell 2024)
  - ▶ Heavy-tailed, but not as heavy as power law
- ▶  $g(d) = \begin{cases} d^\alpha, & d \leq \gamma, \\ \gamma^\alpha + \beta(d - \gamma), & d \geq \gamma \end{cases}$ 
  - ▶ Flexible heavy-tail behaviour



# Modelling recipe

## 1. Regress increment on in-degree

- ▶  $Y_i \sim \text{Poisson} \left( \frac{\mu g(d_i)}{\sum_j g(d_j)} \right)$

## 2. Choice of $g(d)$

- ▶ Piecewise function (previous slide)
- ▶ Power function for benchmark:  $g(d) = d^\alpha$  (for benchmark)
- ▶ Include parameter for zero-appeal  $\delta$

## 3. Infer $(\alpha, \delta, \mu)$ and, if necessary, $(\beta, \gamma)$

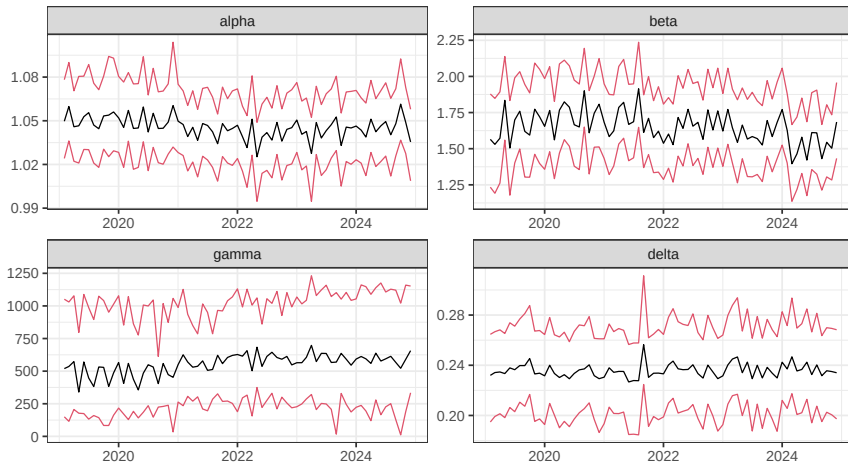
- ▶ Bayesian approach
- ▶ Results of  $\alpha$  most important

# Back to CRAN packages

Depends: R (≥ 3.4.0), [StanHeaders](#) (≥ 2.32.0)  
Imports: methods, stats4, [inline](#) (≥ 0.3.19), [gridExtra](#) (≥ 2.3), [Rcpp](#) (≥ 1.0.7), [RcppParallel](#) (≥ 5.1.4), [loo](#) (≥ 2.4.1), [pkgbuild](#) (≥ 1.2.0), [QuickJSR](#), [ggplot2](#) (≥ 3.3.5)  
LinkingTo: [Rcpp](#) (≥ 1.0.7), [RcppEigen](#) (≥ 0.3.4.0.0), [BH](#) (≥ 1.75.0-0), [StanHeaders](#) (≥ 2.32.0), [RcppParallel](#) (≥ 5.1.4)  
Suggests: [testthat](#) (≥ 3.0.4), parallel, [KernSmooth](#), [shinystan](#), [bayesplot](#), [rmarkdown](#), [rstantools](#), [rstudioapi](#), [Matrix](#), [knitr](#), [coda](#), [V8](#)

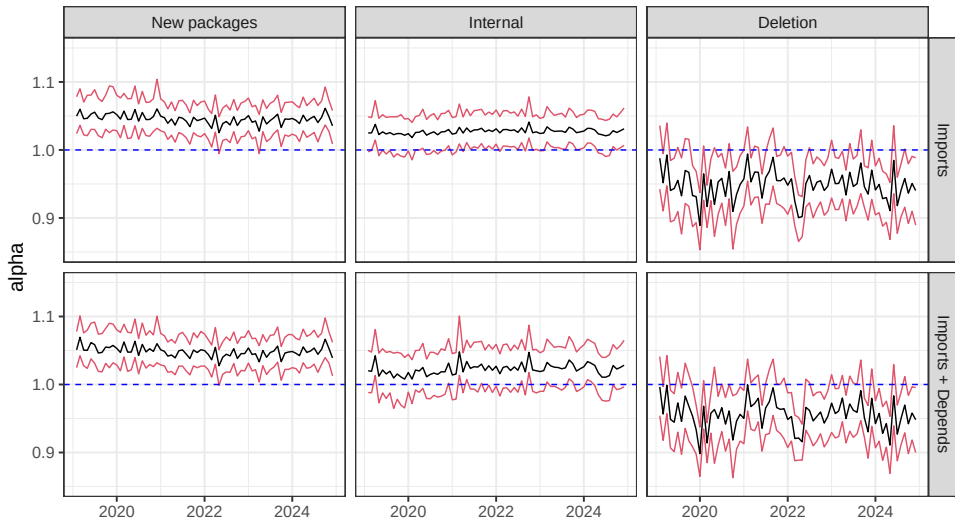
- ▶ **Type of dependencies:** Imports only, as well as Imports + Depends
- ▶ **Subset of data:** New packages, or internal, or deletion
- ▶ **Hierarchical model:** A set of parameters for each month of increments
  - ▶  $(\{\alpha_t\}, \{\beta_t\}, \{\gamma_t\}, \{\delta_t\}, \{\mu_t\})$

# Piecewise function, Imports, new packages

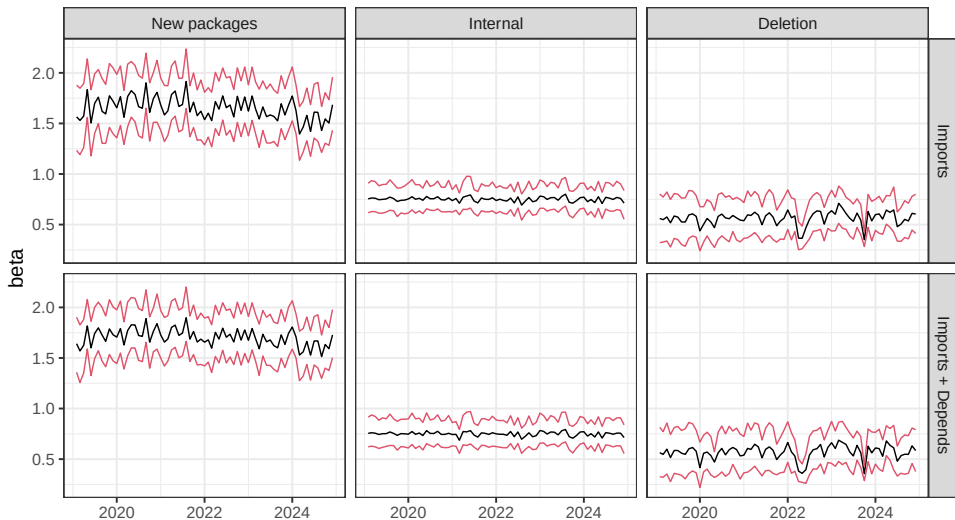


► Very similar results for  $\alpha$  &  $\delta$  using power function

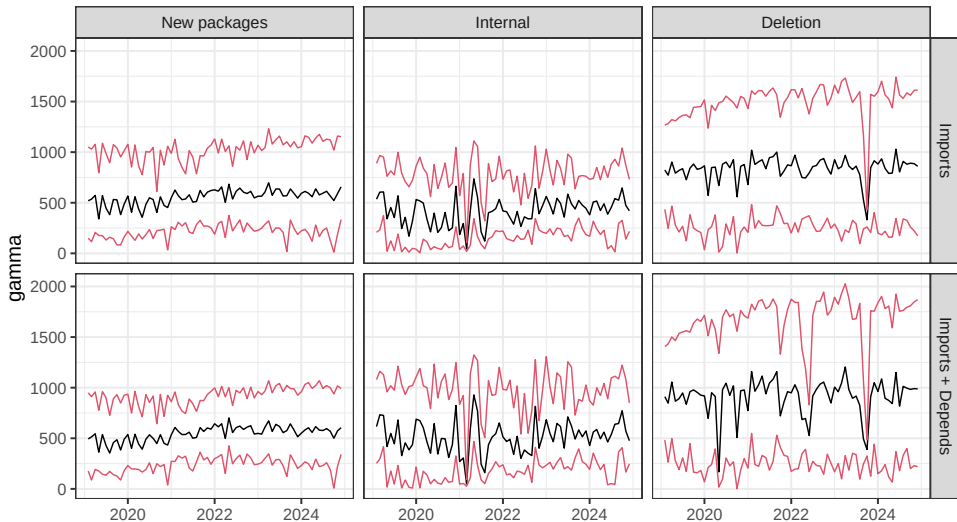
$\alpha$  (super-/sub-linear?)



$\beta$  (slope beyond  $\gamma$ )

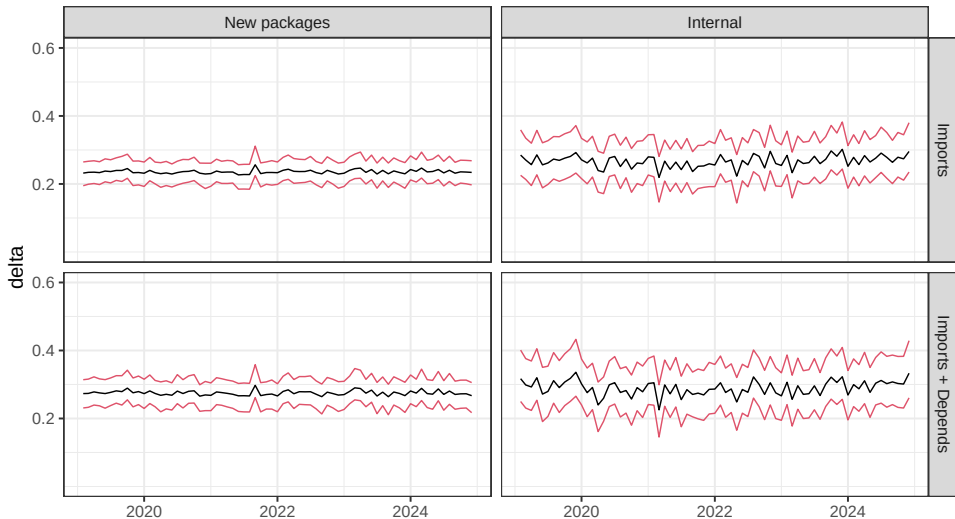


$\gamma$  (piecewise threshold)





$\delta$  (zero appeal)



# Overall

- ▶ Parameters stable over the observation period
- ▶ Adding edges (new packages or internal):
  - ▶ **Partial superlinear preferential attachment (up to  $\gamma$ )**
- ▶ Deleting edges:
  - ▶ **Sublinear preferential detachment without  $\delta$  (zero-appeal)**
- ▶ Not much difference between Imports and Imports + Depends

# Summary

- ▶ Preferential attachment evidenced from increments of network evolution
  - ▶ exploratory visualisations
  - ▶ Poisson regression model
- ▶ Piecewise weight function of in-degree
  - ▶ accommodates flexible heavy tail (of the degree distribution)
  - ▶ reveals partial super-/sub-linear preferential attachment
- ▶ Model applied to CRAN package dependencies successfully
- ▶ Next step: Model selection between power & piecewise functions

# Thank you

- ▶ This presentation: <https://bit.ly/sunbelt2025>
- ▶ Paper on partial scale-freeness: <https://doi.org/10.1111/stan.12355>



## Bibliography

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